

Number Series

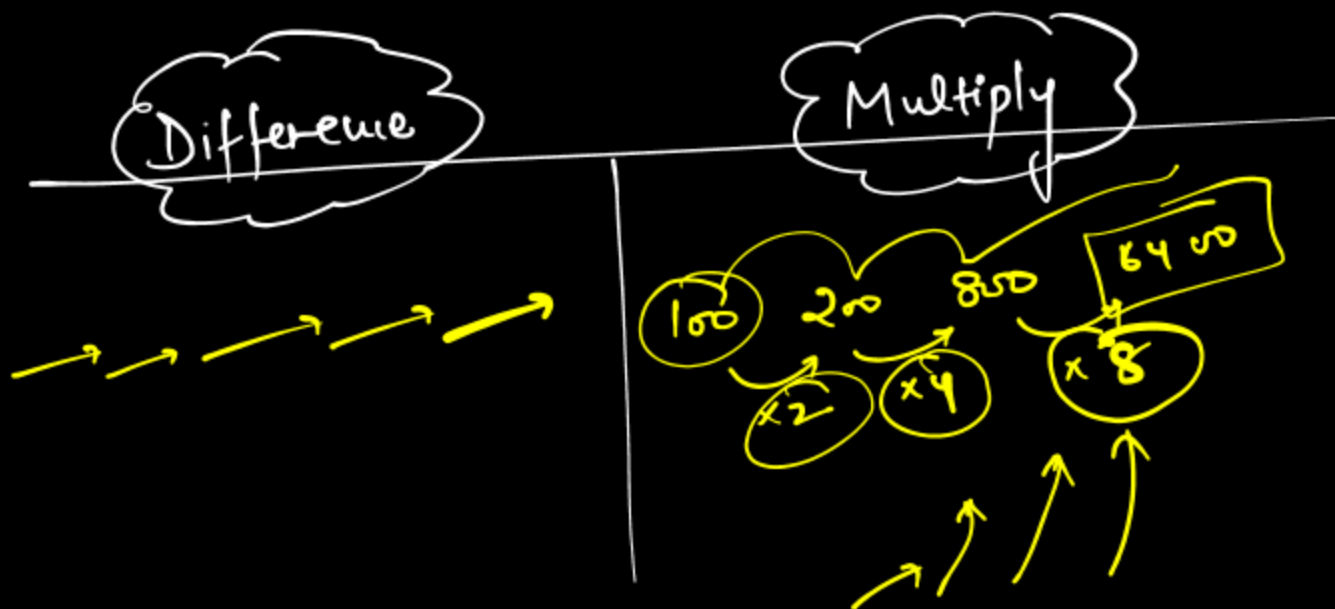
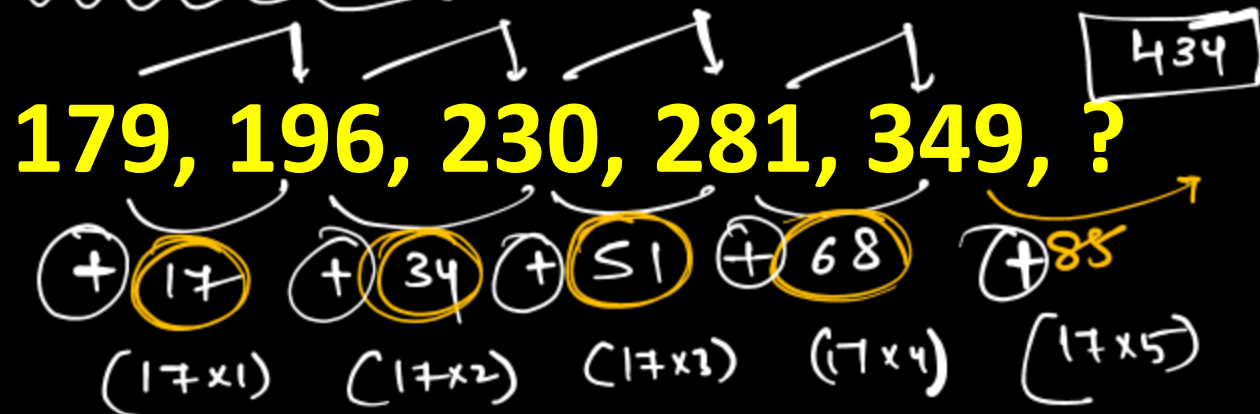
[Understanding All Patterns]

By Aashish Arora

Tables ✓

Difference Based (Consecutive multiples addition)

Difference Based



Practice based on Similar Pattern

438, 461, 507, 576, 668, 783 ✓

+23 +46 +69 +92 +115

↑ ↑ ↑ ↑

337, 432, 546, 679, 831, 1002 ✓

+95 +114 +133 +152 +171

(19x5) (19x6) (19x7) (19x8) (19x9)

114 95 76 57 38

Difference Based (Consecutive multiples subtraction)

772, 734, 677, 601, 506, ?

$\begin{array}{ccccc} \text{---} \nearrow & \text{---} \nearrow & \text{---} \nearrow & \text{---} \nearrow & \text{---} \nearrow \\ \textcircled{-38} & \textcircled{-57} & \textcircled{-76} & \textcircled{-95} & \textcircled{-114} \\ (19 \times 2) & (19 \times 3) & (19 \times 4) & (19 \times 5) & 19 \times 6 \end{array}$

$\textcircled{392}$ ✓

Practice based on Similar Pattern

941, 824, 720, 629, 551, 486

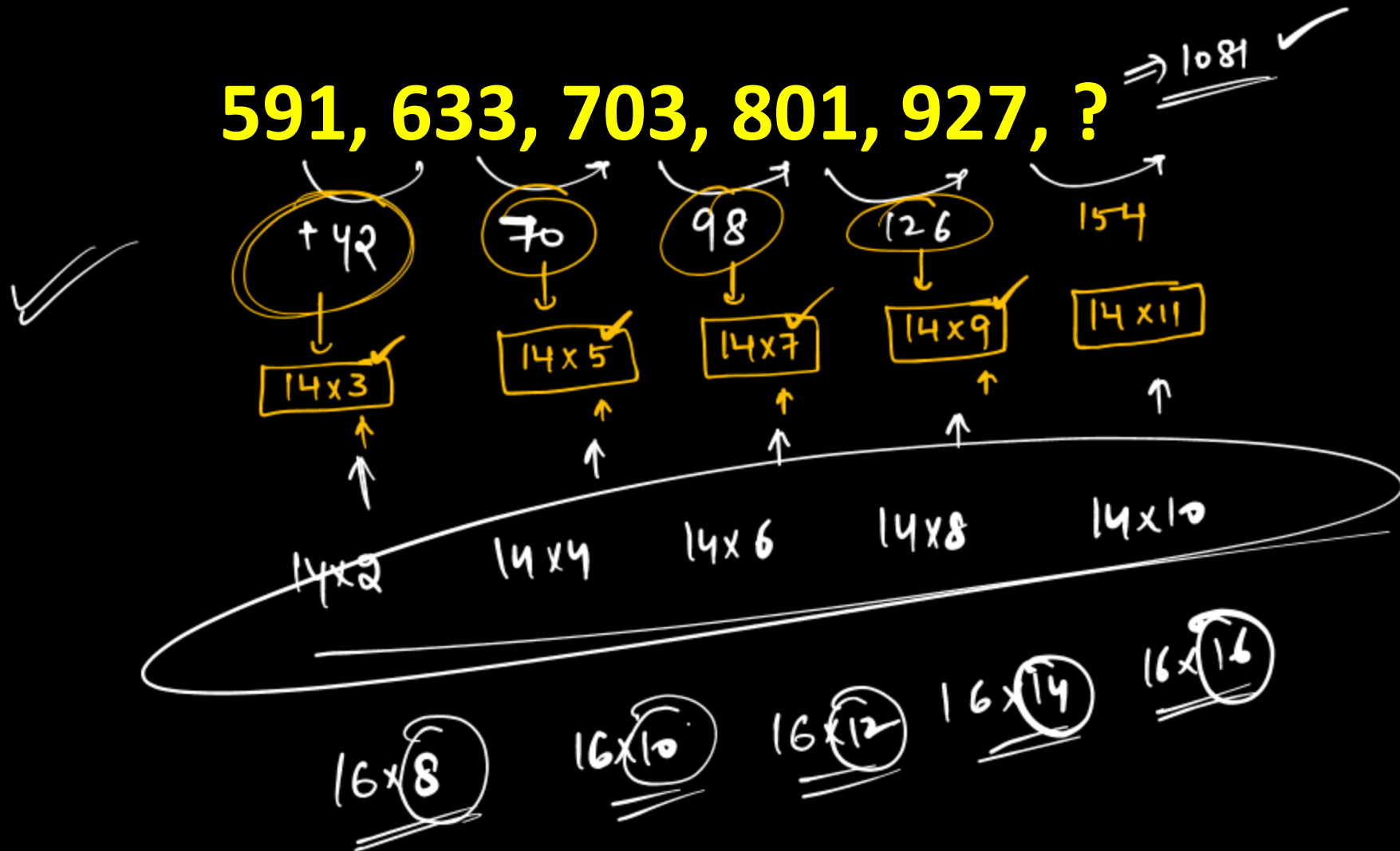
-117 -104 -91 -78 -65

648, 619, 561, 474, 358, 213

-29 -58 -87 -116 -145

Difference Based Tables (Alternate multiples addition)

591, 633, 703, 801, 927, ? $\Rightarrow 1081$ ✓



Practice based on Similar Pattern

193, 211, 265, 355, 481, 643 ✓

18 54 90 126 162

18 × 1 18 × 3 18 × 5 18 × 7 18 × 9

49, 170, 313, 478, 665, 874 ✓

121 143 165 187 209

11 × 11 11 × 13 11 × 15 11 × 17 11 × 19

Difference Based Tables (Alternate multiples subtraction)

891, 811, 699, 555, 379, ?

$$\begin{array}{r} \text{---} 80 \\ \text{---} 112 \\ \text{---} 144 \\ \text{---} 176 \\ \text{---} 208 \end{array}$$

$$\begin{array}{r} (16 \times 5) \\ (16 \times 7) \\ (16 \times 9) \\ (16 \times 11) \\ (16 \times 13) \end{array}$$

$$(171)$$

$$\begin{array}{r} 21 \\ 35 \\ \hline 76 \end{array}$$

Practice based on Similar Pattern

563, 536, 491, 428, 347, 248

$$\begin{array}{ccccc} \text{---} & \text{---} & \text{---} & \text{---} & \text{---} \\ \text{---} & \text{---} & \text{---} & \text{---} & \text{---} \\ -27 & -45 & -63 & -81 & -99 \\ (9 \times 3) & (9 \times 5) & (9 \times 7) & (9 \times 9) & (9 \times 11) \end{array}$$

711, 692, 635, 540, 407, 236

$$\begin{array}{ccccc} \text{---} & \text{---} & \text{---} & \text{---} & \text{---} \\ \text{---} & \text{---} & \text{---} & \text{---} & \text{---} \\ -19 & -57 & -95 & -133 & -171 \end{array}$$

Prime No.
अभाज्य संख्याएं

Nos. which are divisible by 1 and itself.

Two factors $\Rightarrow$ 1, itself

2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59,
61, 67, 71, 73, 79, 83, 89, 97, ...

1 to 100

25 prime nos

31 37 41 43 47

Difference Based Tables (Prime multiples addition)

68, 114, 183, 298, 459, ?

$+46$ $+69$ $+115$ $+161$ $+253$
 23×2 23×3 23×5 23×7 23×11

712 ✓

Practice based on Similar Pattern

37, 71, 122, 207, 326, 573

$+34$
 17×2

$+51$
 17×3

$+85$
 17×5

$+119$
 17×7

$+187$
 17×11

94, 133, 198, 289, 432, 601

$+39$
 13×3

$+65$
 13×5

$+91$
 13×7

$+143$
 13×11

$+169$
 13×13

Gap of Squares

283, 404, 548, 717, 913, ?

+121 +144 +169 +196 +225

11^2 12^2 13^2 14^2 15^2

1138 ✓

Practice based on Similar Pattern

261, 297, 361, 461, 605, 801

even

36
6²

81
9²

36
6²

36
6²

00
0²

00
0²

537, 586, 667, 788, 957, _____

odd

9
3²

64
8²

36
6²

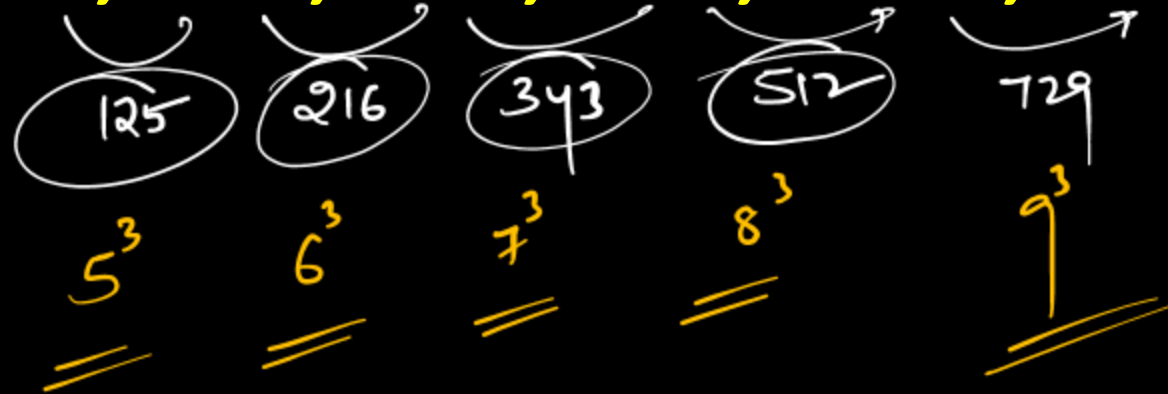
64
8²

25
5²

#

Gap of Cubes

87, 212, 428, 771, 1283, ?



✓✓
2012



Practice based on Similar Pattern

603, 595, 568, 504, 379, 163

$$\begin{array}{ccccc} \curvearrowright & \curvearrowright & \curvearrowright & \curvearrowright & \curvearrowright \\ -8 & -27 & -64 & -125 & -216 \\ \hline & & & & \end{array}$$

71, 287, 799, 1799, 3527, 6271

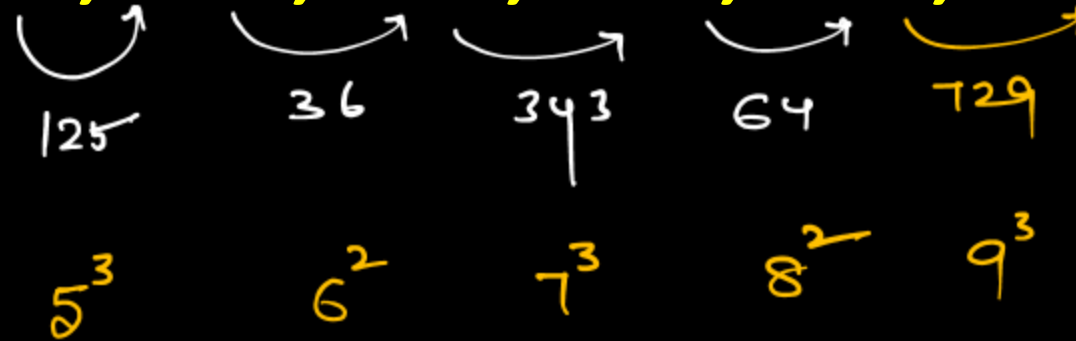
$$\begin{array}{ccccc} \curvearrowright & \curvearrowright & \curvearrowright & \curvearrowright & \curvearrowright \\ +216 & +512 & +1000 & 1728 & 2744 \\ \hline 6^3 & 8^3 & 10^3 & 12^3 & 14^3 \end{array}$$

RRB

#

Gap of Squares & Cubes (mix)

117, 242, 278, 621, 685, ?



1414

✓

Practice based on Similar Pattern

✓✓ 238, 319, 1319, 1440, 3168, 3337

$\begin{array}{ccccc} \text{81} & \text{1000} & \text{121} & \text{1728} & \text{169} \\ \text{9}^2 & \text{10}^3 & \text{11}^2 & \text{12}^3 & \text{13}^2 \end{array}$

✓✓ 6130, 5905, 3161, 2992, 1264, 1143

$\begin{array}{ccccc} -225 & -2744 & -169 & -1728 & -121 \\ \text{15}^2 & \text{14}^3 & \text{13}^2 & \text{12}^3 & \text{11}^2 \end{array}$

Practice based on Similar Pattern



$$484, 493, 429, 454, 238, \underline{287}$$

$+9$	-64	$+25$	-216	$+49$
3^2	4^3	5^2	6^3	7^2

Gap of power's of Similar or Diff. No.

39, 40, 44, 71, 327, ?

1 4 27 256 3125

1^1

2^2

3^3

4^4

5^5

$[4^2]^2$

4^4

3452 ✓

Practice based on Similar Pattern

68, 69, 85, 112, 128, 133 ✓

$+1$ $+16$ $+27$ $+16$ $+5$

$\downarrow$ $\downarrow$ $\downarrow$ $\downarrow$ $\downarrow$

5 4 3 2 1

1 2 3 4 5

Double difference Based

345, 382, 431, 494, 573, ?

$$\begin{array}{ccccccccc} & \underbrace{} & & \underbrace{} & & \underbrace{} & & \underbrace{} & & \underbrace{} \\ + & 37 & & + & 49 & & + & 63 & & + & 79 & & + & 97 \\ & \underbrace{} & & \underbrace{} & & \underbrace{} & & \underbrace{} & & & & & & \\ & + & 12 & & + & 14 & & + & 16 & & + & 18 & & \end{array}$$

670 ✓

Practice based on Similar Pattern

116, 144, 182, 235, 308, 406

28 38 53 73 98

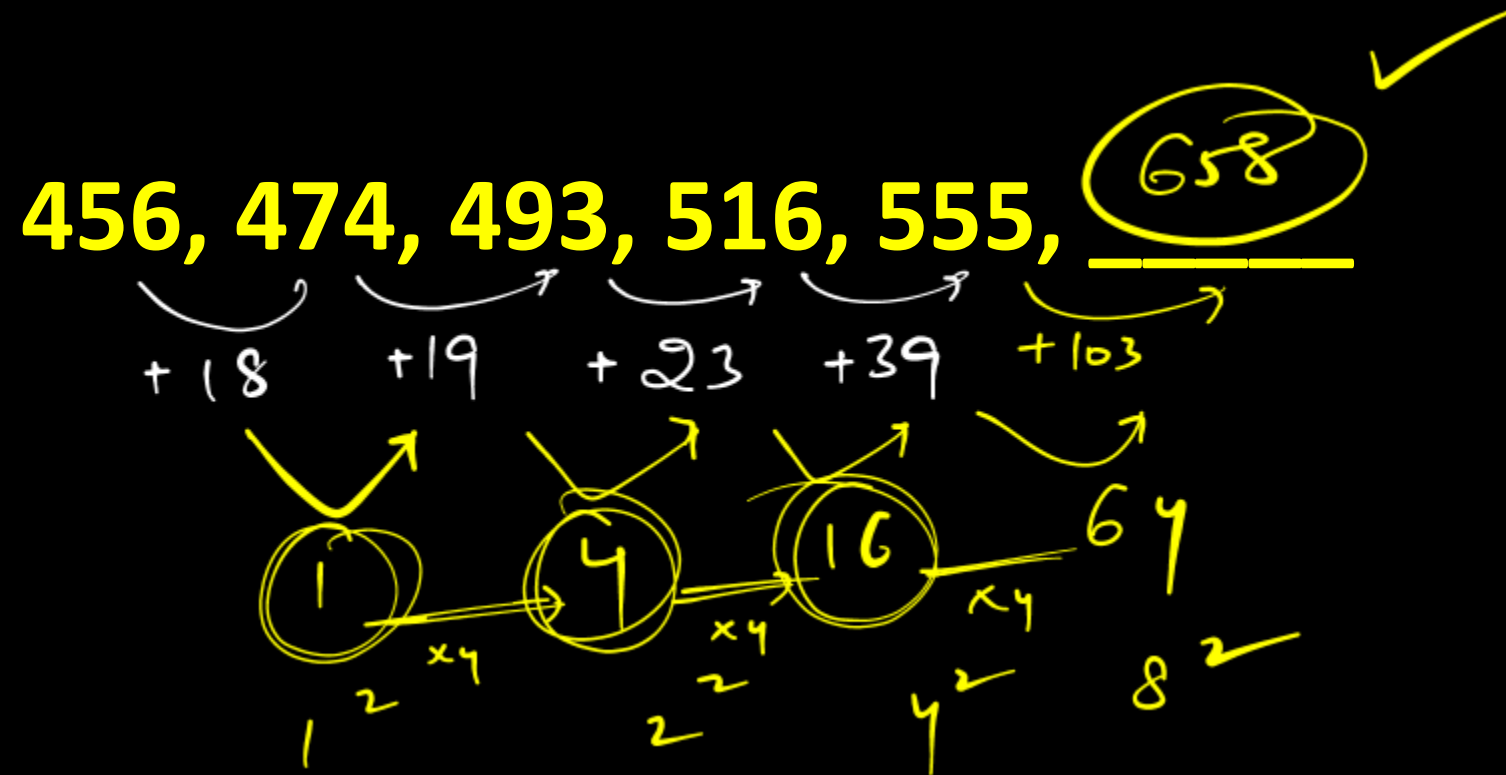
+10 +15 +20 +25

56, 188, 319, 442, 538, 570

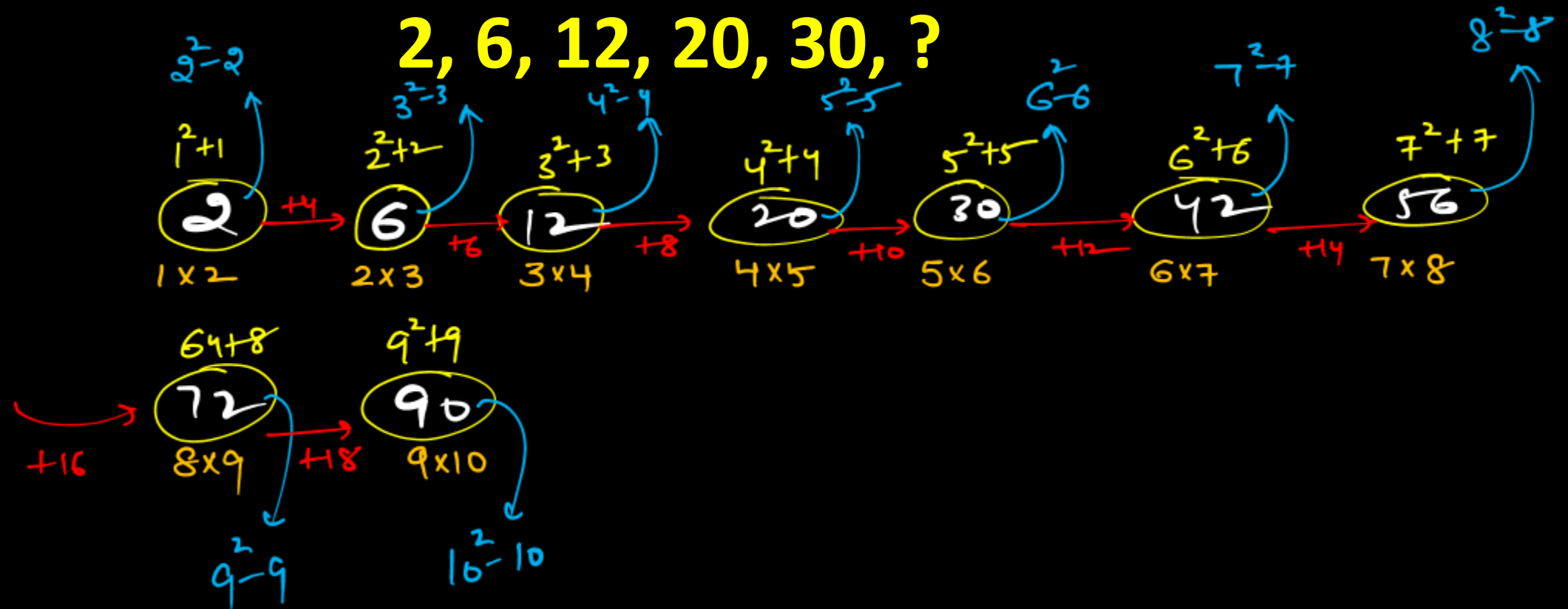
+132 +131 +123 +96 +32

-1 -8 -27 -64

Practice based on Similar Pattern



Frequently Used Patterns



Things To Remember

$$1 * 2 = 2$$

$$2 * 3 = 6$$

$$3 * 4 = 12$$

$$4 * 5 = 20$$

$$5 * 6 = 30$$

$$6 * 7 = 42$$

$$7 * 8 = 56$$

$$8 * 9 = 72$$

$$9 * 10 = 90$$

$$10 * 11 = 110$$

$$11 * 12 = 132$$

$$12 * 13 = 156$$

$$13 * 14 = 182$$

$$14 * 15 = 210$$

$$15 * 16 = 240$$

$$16 * 17 = 272$$

$$17 * 18 = 306$$

$$18 * 19 = 342$$

$$19 * 20 = 380$$

$$20 * 21 = 420$$

$$21 * 22 = 462$$

Practice based on Similar Pattern

12×13 13×14 14×15 15×16 16×17 17×18
156, 182, 210, 240, 272, 306

2, 12, 30, 56, 90, 132

$\checkmark \downarrow \checkmark$
 1×2

$\checkmark \downarrow \checkmark$
 3×4

$\checkmark \downarrow \checkmark$
 5×6

$\checkmark \downarrow \checkmark$
 7×8

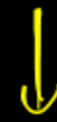
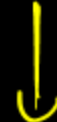
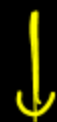
$\checkmark \downarrow \checkmark$
 9×10

11×12

Frequently Used Patterns

$$\text{Pattern} = a^3 + a^2$$

2, 12, 36, 80, 150, ?



252

$$1^3 + 1^2 = 2$$

$$2^3 + 2^2 = 12$$

$$3^3 + 3^2 = 36$$

$$4^3 + 4^2 = 80$$

$$5^3 + 5^2 = 150$$

$$6^3 + 6^2 = 252$$



Frequently Used Patterns

$$\text{Pattern} = a^3 - a^2$$

0, 4, 18, 48, 100, ?

180

$$1^3 - 1^2 = 0$$

$$2^3 - 2^2 = 4$$

$$3^3 - 3^2 = 18$$

$$4^3 - 4^2 = 48$$

$$5^3 - 5^2 = 100$$

$$6^3 - 6^2 = 180$$

Twisted Square and cubes Pattern

479, 486, 512, 575, 699, ?

$$\begin{array}{cccccc} & \text{+7} & \text{+26} & \text{+63} & \text{+124} & \text{+215} \\ & 2^3-1 & 3^3-1 & 4^3-1 & 5^3-1 & 6^3-1 \end{array}$$

914

$$3^3-1$$

Practice based on Similar Pattern

#

152, 217, 297, 398, 518, 663 ✓

$$\begin{array}{c} \text{152} \\ \text{65} \end{array}$$

$$\begin{array}{c} \text{217} \\ \text{80} \end{array}$$

$$\begin{array}{c} \text{297} \\ \text{101} \end{array}$$

$$\begin{array}{c} \text{398} \\ \text{120} \end{array}$$

$$\begin{array}{c} \text{518} \\ \text{145} \end{array}$$

$$8^2 + 1$$

$$9^2 - 1$$

$$10^2 + 1$$

$$11^2 - 1$$

$$12^2 + 1$$

Practice based on Similar Pattern

86, 95, 105, 170, 196, _____

$\underbrace{\hspace{1.5cm}}$	$\underbrace{\hspace{1.5cm}}$	$\underbrace{\hspace{1.5cm}}$	$\underbrace{\hspace{1.5cm}}$	$\underbrace{\hspace{1.5cm}}$
9	10	65	26	217
$\downarrow$		$\downarrow$	$\downarrow$	
$2^3 + 1$	$3^2 + 1$	$4^3 + 1$	$5^2 + 1$	$6^3 + 1$

413 ✓